Fish-based / marine-protein pet diets: HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified human-grade standard for fish-based / marine-protein pet diets, stated as consumed. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

1 | How every number on this standard is set

Pet food is legally governed by feed law — EU Directive 2002/32/EC on undesirable substances in animal feed, FDA Center for Veterinary Medicine guidance, and AAFCO model regulations — whose maximum levels are generally laxer than the corresponding human-food limits. Japan (FAMIC pet-food standard) and China (draft pet-food guidance) set enforced or proposed pet-food maxima in the same lax range. The HMTc standard for each analyte is not the feed limit: it is the strictest applicable human-food sovereign maximum for the nearest human-food analog, adopted as a voluntary human-grade over-compliance. The feed floor is disclosed on every row and never selects the number.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS

Lead (Pb)	200	read-across (FJ)
Cadmium (Cd)	50	read-across (GSO)
Total arsenic (tAs)	1,000	read-across (Paraguay)
Inorganic arsenic (iAs)	100	read-across (European Union)
Mercury, total (tHg)	300	read-across (EU 2023/915)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	800	read-across (EU 2024/1987)
Tin (Sn)	200,000	read-across (EU 2023/915)
Aluminium (Al)	100,000	read-across (China)
Total chromium (Cr)	1,000	read-across (Hong Kong)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	200	read-across (FJ); Muscle meat of fish, excluding those listed below
Cadmium (Cd)	50	read-across (GSO); Muscle meat of fish with exception mentioned
Total arsenic (tAs)	1,000	read-across (Paraguay); raw, frozen, or refrigerated fish
Inorganic arsenic (iAs)	100	read-across (European Union); muscle meat of fish, species other than those listed under Annex I 3.4.5.2
Mercury, total (tHg)	300	read-across (EU 2023/915); listed lower-mercury fish and cephalopods/marine gastropods, including cod, herring, mackerel, salmon/trout species listed, sardine/pilchard, sole, whiting, and others
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)

Nickel (Ni)	800	read-across (EU 2024/1987); cereals except durum/rice/husked-rice/-pseudocereal/oat rows
Tin (Sn)	200,000	read-across (EU 2023/915); canned foods other than canned beverages, except canned dried and canned powdered products
Aluminium (Al)	100,000	read-across (China); baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder
Total chromium (Cr)	1,000	read-across (Hong Kong); Fish
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

4 | Cite this standard

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References

[1] FJ
Binds Pb.
READ-ACROSS

[2] GSO
Binds Cd.
READ-ACROSS

[3] Paraguay
Binds tAs.
READ-ACROSS

[4] EU 2023/915
Binds tHg, Sn.
READ-ACROSS

[5] EU 2024/1987
Binds Ni. cereals except durum/rice/husked-rice/pseudocereal/oat rows.
READ-ACROSS

[6] Hong Kong
Binds Cr.
READ-ACROSS

[7] China
Binds Al. baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder.
READ-ACROSS

[8] European Union
Binds iAs.
READ-ACROSS

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/pet-food-fish-marine-protein. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/pet-food-fish-marine-protein.